

Topic 3: Stoichiometry - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Moles, formulae, equations and chemical calculations

Questions in this answer PDF: 17

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2021 May/June Paper 42 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 7 | Answer source: 0620_s21_ms_42.pdf

Answer points

3(a)

- (thermal) decomposition

3(b)

- 84 (1)
- $12.6 / 84 = 0.15(00)$ (1)
- $0.15(00) / 2 = 0.075(00)$ (1)
- $0.075(0) \times 24.0 = 1.8$ (1)

3(c)

- $\text{Ca}(\text{OH})_2$ (1)
- CaCO_3 (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Oct/Nov Paper 43 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 16 | Answer source: 0620_w21_ms_43.pdf

Extra topic(s): Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds, Topic 10: Chemistry of the environment

Answer points

3(a)

3(b)

- $(207 / 239 \times 100 =)$ 86.6%

3(c)

- PbS because the percentage of lead is larger in PbS or answer to (b) > 77.5%

3(d)

- $\text{PbCO}_3 \rightarrow \text{PbO} + \text{CO}_2$

3(e)

- $\text{PbCO}_3 + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$

3(f)(i)

- two double bonds (1)
- two pairs of non-bonding electrons on each oxygen and no non-bonding electrons on carbon (1)

3(f)(ii)

- bonds between ions or ionic bonds in lead(II) oxide (1)
- attraction between molecules in carbon dioxide (1)
- weaker attraction (between particles) in carbon dioxide opposite reasoning also allowed (1)

3(g)

- $\text{Mg} + \text{Pb}^{2+} \rightarrow \text{Mg}^{2+} + \text{Pb}$ (1)
- no reaction (1)

3(h)(i)

- glowing splint (1)
- relights (1)

3(h)(ii)

- nitrogen dioxide /
- nitrogen(IV) oxide

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Feb/March Paper 42 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 15 | Answer source: 0620_m22_ms_42.pdf

Extra topic(s): Topic 10: Chemistry of the environment, Topic 11: Organic chemistry

Answer points

3(a)

- nitrogen (from air) and oxygen (from air) react
- react due to high temperatures (of engine)

3(b)

- acid rain

3(c)(i)

- (thermal) decomposition

3(c)(ii)

- HCNO

3(c)(iii)

- (damp red) litmus
- (litmus) turns blue

3(d)(i)

- $6\text{NO}_2 + 8\text{NH}_3 \rightarrow 7\text{N}_2$
- either 6NO_2 or 8NH_3
- all three balanced

3(d)(ii)

- (nitrogen) loses oxygen

3(d)(iii)

- reduction and oxidation occur

3(e)

- Mr urea = 60
- $135 \times 60 = 8100$ and g to kg conversion = 8.1(00) kg

3(f)(i)

- carbon monoxide

3(f)(ii)

- carbon dioxide and nitrogen
- February/March 2022

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 May/June Paper 41 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 12 | Answer source: 0620_s22_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis

Answer points

3(a)(i)

- positive ions / cations (1)
- sea of electrons / mobile electrons / delocalised electrons (1)
- attraction between positive ions and electrons (1)

3(a)(ii)

- electrons move /
- electrons mobile /
- electrons flow

3(b)(i)

- ionic

3(b)(ii)

- ions move /
- ions are mobile /
- ions flow

3(c)(i)

- pipette

3(c)(ii)

- yellow to orange

3(c)(iii)

- at least two results are within 0.2 cm³ or less

3(c)(iv)

- 0.005 / 5 → 10⁻³ (1)
- 0.0025 / 2.5 → 10⁻³ (1)
- 0.125 (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2022 May/June Paper 42 Question 2

Main topic: Topic 3: Stoichiometry | Marks: 21 | Answer source: 0620_s22_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points**2(a)(i)**

- $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{H}_2$
- H₂ as product
- Fully correct equation

2(a)(ii)

- calcium oxide

2(b)(i)

- 7 < pH < 12

2(b)(ii)

- OH⁻

2(c)(i)

- carbon dioxide

2(c)(ii)

- (a solution that) can dissolve no more solute
- at a given temperature

2(c)(iii)

- add excess (solid) calcium hydroxide (to water)
- filter

2(c)(iv)

- (aqueous) sodium hydroxide
- white ppt
- insoluble / remains in excess

2(d)(i)

- burette

2(d)(ii)

- neutralisation

2(d)(iii)

- indicator

2(d)(iv)

- Mol HCl = 0.0500 → 20.0 / 1000 = 0.001(00)
- Mol Ca(OH)₂ = / 2 = 0.00100 / 2 = 0.0005(00)
- → 1000 / 25 = 0.0005(00) → 40 = 0.02(00)
- Mr Ca(OH)₂ = 74
- → = 74 → 0.02 = 1.48 (g / dm³)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2022 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 14 | Answer source: 0620_w22_ms_41.pdf

Extra topic(s): Topic 12: Experimental techniques and chemical analysis, Topic 7: Acids, bases and salts

Answer points

4(a)

- $\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$
- H_2O and CO_2 as product (1)
- rest of equation correct (1)

4(b)

- fizzing / effervescence (1)
- solid disappears / dissolves (1)

4(c)

- filtrate

4(d)(i)

- a solution that can contain no more solute (1)
- at a given temperature (1)

4(d)(ii)

- cool the solution

4(e)(i)

- anhydrous

4(e)(ii)

- Mr $\text{Ca}(\text{NO}_3)_2 = 164$ (1)
- mol $\text{Ca}(\text{NO}_3)_2 = 2.46 / 164 = 0.015(00)$ (1)
- $0.015(00) / 0.015(00) = 1$
- $0.0600 / 0.015(00) = 4$ and $x = 4$ (1)

4(f)

- $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
- NaNO_2 on the right-hand side
- equation completely correct

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2022 Oct/Nov Paper 42 Question 2

Main topic: Topic 3: Stoichiometry | Marks: 22 | Answer source: 0620_w22_ms_42.pdf

Answer points

2(a)(i)

- to prevent contact with air / oxygen and / or water

2(b)(i)

- combustion

2(b)(ii)

- yellow

2(b)(iii)

- $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$
- species (1)
- balancing (1)

2(b)(iv)

- eight crosses in second shell of both Na (1)
- six dots and two crosses in second shell of O (1)
- '+' charge on each Na ion on correct answer line and '2-' charge on O ion on correct answer line (1)

2(c)(i)

- proton acceptor

2(c)(ii)

- pH 14

2(c)(iii)

- yellow

2(c)(iv)

- $\text{mol Na} = 0.345 / 23 = 0.015(00)$ (1)

- $\text{mol NaOH} = 0.015(00)$ (1)

- $\rightarrow 1000 / 50 = 0.015(00) \rightarrow 20 = 0.3(00)$ (1)

- $\text{Mr NaOH} = 40(1)$

- $\rightarrow = 40 \times 0.3 = 12.0$ (g / dm³) (1)

2(d)(i)

- precipitation

2(d)(ii)

- red-brown

2(d)(iii)

- iron(III) hydroxide

2(d)(iv)

- $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$

- $\text{Fe}(\text{OH})_3$ (as only product) (1)

- Fe^{3+} and 3OH^{-} (as only reactants) (1)

- state symbols (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Oct/Nov Paper 42 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 12 | Answer source: 0620_w22_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis

Answer points

4(a)

- $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$

4(b)

- no (more) fizzing (1)

- solid stops dissolving or a solid remains / is visible (in the mixture) (1)

4(c)

- residue

4(d)(i)

- saturated

4(d)(ii)

- solubility (of MgSO_4 / solid) decreases (as temperature decreases)

4(e)(i)

- hydrated

4(e)(ii)

- $\text{Mr MgSO}_4 = 120$ (1)

- $\text{mol MgSO}_4 = 2.40 / 120 = 0.02(00)$ (1)

- $0.02(00) / 0.02(00) = 1$

- $0.140 / 0.02(00) = 7$ and $x = 7$ (1)

4(f)

- $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$

- $\text{NO}_2 + \text{O}_2$ (1)

- equation completely correct (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 12 | Answer source: 0620_w23_ms_41.pdf

Answer points

4(a)

- test: relights
- AND
- observations: a glowing splint

4(b)(i)

- lower gradient (at t₂)

4(b)(ii)

- concentration (of H₂O₂ particles) decreases (1)
- frequency of collisions between particles decreases (1)

4(b)(iii)

- steeper curve which does not cross original curve and levels off before the original curve (1)
- finishes at same volume (1)

4(c)

- mol O₂ = $72 / 24000 = 0.003(00)$ (1)
- mol H₂O₂ = $\rightarrow 2 = 0.002 \rightarrow 2 = 0.006(00)$ (1)
- conc H₂O₂ = $\rightarrow 1000 / 20 = 0.006 \rightarrow 1000 / 20 = 0.3(00)$ (1)
- Mr H₂O₂ = 34 (1)
- $= \rightarrow = 34 \rightarrow 0.3 = 10.2$ (g / dm³) (1)

4(d)

- any transition metal oxide

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2024 May/June Paper 41 Question 5

Main topic: Topic 3: Stoichiometry | Marks: 17 | Answer source: 0620_s24_ms_41.pdf

Extra topic(s): Topic 4: Electrochemistry, Topic 7: Acids, bases and salts

Answer points

5(a)(i)

- lead(II) nitrate(1)
- sodium bromide /
- potassium bromide /
- ammonium bromide(1)

5(a)(ii)

- filter precipitate or lead(II) bromide or solid or residue(1)
- wash the residue with distilled water AND dry e.g. between filter papers(1)

5(a)(iii)

- $\text{Pb}^{2+}(\text{aq}) + 2\text{Br}^{-}(\text{aq}) \rightarrow \text{PbBr}_2(\text{s})$
- PbBr₂ as the only product (1)
- $\text{Pb}^{2+} + 2\text{Br}^{-}$ as the only reactants in a balanced equation(1)
- state symbols
- $(\text{aq}) + (\text{aq}) \rightarrow (\text{s})(1)$

5(b)

- acidified aqueous potassium manganate(VII) (1)
- purple to colourless (1)

5(c)

- $2\text{CoO} + 4\text{NO}_2$ (1)
- $8\text{H}_2\text{O}$ (1)

5(d)(i)

- heat again and weigh again
- OR
- repeat steps 2 and 3(1)
- until mass is constant (1)

5(d)(ii)

- $0.005 / 5 \rightarrow 10^{-3}$ (1)

- 0.63 (1)
- $(0.63 / 18 \Rightarrow) 0.035$ (1)
- $(0.035 \div 0.005 \Rightarrow) 7$ (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 42 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 15 | Answer source: 0620_s24_ms_42.pdf

Extra topic(s): Topic 5: Chemical energetics

Answer points

3(a)

- Haber

3(b)

- enthalpy change (of reaction)

3(c)

- 450 and $^{\circ}\text{C}$
- 200 and atm or 20 000 and kPa
- iron

3(d)

- increase temperature
- increase pressure

3(e)

- bond energy in breaking bonds
- $= [945 + (3 \rightarrow 435)] = (+) 2250$ (kJ / mol)
- $= 2250 + 90 = 2340$
- $= 2340 / 6 = 390$

3(f)(i)

- H_2O

3(f)(ii)

- Ammonium chloride

3(f)(iii)

- mol $\text{CaO} = 1.12 / 56 = 0.02(00)$
- mol $\text{NH}_3 = \rightarrow 2 = 0.02(00) \rightarrow 2 = 0.04(00)$
- vol $\text{NH}_3 = \rightarrow 24000 = 0.04 \rightarrow 24000 = 960$

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.*

2024 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 12 | Answer source: 0620_w24_ms_41.pdf

Answer points

4(a)(i)

- +5
- +4

4(a)(ii)

- oxygen

4(a)(iii)

- Mr of $\text{Mg}(\text{NO}_3)_2 = 148$
- mol of $\text{Mg}(\text{NO}_3)_2 = 7.40 / 148 = 0.05(00)$ (mol)
- mol of $\text{NO}_2 = 0.05(00) \rightarrow 4 / 2 = 0.1(00)$ (mol)
- volume of $\text{NO}_2 = 0.1(00) \rightarrow 24\,000 = 2400$ (cm³)

4(b)

- eight crosses in second shell of Mg
- six dots and two crosses in second shell of O
- '2+' charge on Mg ion on correct answer line AND '2-' charge on O ion on correct answer line

4(c)

- two dot-cross pairs as a double bond
- four non-bonding electrons on each O

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

2024 Oct/Nov Paper 42 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 14 | Answer source: 0620_w24_ms_42.pdf

Answer points**4(a)**

- $\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$

4(b)

- (silver) nitrate

4(c)

- $0.01 \rightarrow 1000 / 50 = 0.2(00) \text{ (mol / dm}^3\text{)}$

4(d)

- cream

4(e)

- mol of AgBr = $0.01(00)$
- Mr of AgBr = 188
- $188 \times 0.01 = 1.88 \text{ (g)}$

4(f)

- sodium ethanoate

4(g)(i)

- rinsing of residue

4(g)(ii)

- (crystals of) sodium ethanoate

4(h)(i)

- named soluble barium salt e.g. barium chloride / barium nitrate
- named soluble sulfate salt e.g. sodium sulfate / potassium sulfate / ammonium sulfate / sulfuric acid

4(h)(ii)

- any equation with BaSO_4 as a product
- correct equation using soluble reactants

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 May/June Paper 41 Question 4

Main topic: Topic 3: Stoichiometry | Marks: 15 | Answer source: 0620_s25_ms_41.pdf

Extra topic(s): Topic 10: Chemistry of the environment, Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds

Answer points**4(a)**

- 3 dots and 3 crosses shared
- one non-bonding pair of 2 dots on one N atom and 2 crosses on the other N atom to complete the octet on both

4(b)(i)

- Conditions can be in any order
- (temperature) 450 and $^{\circ}\text{C}$
- (pressure)
- 200 and atmospheres / 20 000 and kPa
- iron and catalyst

4(b)(ii)

- $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

4(c)(i)

- $5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$

4(c)(ii)

- -3

- +2

4(c)(iii)

- increase in oxidation number

4(c)(iv)

- $4\text{NO} + 3\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$

- $\text{HNO}_3(1)$

- equation completely correct

4(d)

- 15

- 7.5

- 990

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2025 Oct/Nov Paper 41 Question 2

Main topic: Topic 3: Stoichiometry | Marks: 17 | Answer source: 0620_w25_ms_41.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

2(a)

- strontium / Sr

2(b)

- metallic

- positive ions / cations

- sea of electrons / mobile electrons / delocalised electrons

- attraction between positive ions and electrons

2(c)(i)

- effervescence

- solid dissolves

- solution turns blue

2(c)(ii)

- calcium hydroxide (1)

- hydrogen (1)

2(d)(i)

- orange-red

2(d)(ii)

- $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$

- CaO as only product

- correct equation

2(e)(i)

- water(s) of crystallisation

2(e)(ii)

- $3.28 \times 164 = 0.02(00)$

- $1.44 \times 18 = 0.08(00)$

- $\quad = 4$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 41 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 14 | Answer source: 0620_w25_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

3(a)

- $2\text{H}_2\text{O}$

- state symbols (aq) and (l)

3(b)

- neutralisation

3(c)

- $0.0100 \rightarrow 1000 \times 40 = 0.25(0)$

3(d)

- burette

3(e)

- mol H_2SO_4 needed = $0.0100 \times 2 = 0.005(00)$

- vol of H_2SO_4 needed = $0.005(00) \times 0.200 \rightarrow 1000 = 25 \text{ (cm}^3\text{)}$

3(f)

- yellow

- orange

3(g)

- repeat steps 1 and 3 without indicator

3(h)(i)

- (a solution containing the) maximum concentration of a solute dissolved in the solvent

- at a specified temperature

3(h)(ii)

- solubility decreases as temperature decreases

3(h)(iii)

- none

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 42 Question 3

Main topic: Topic 3: Stoichiometry | Marks: 13 | Answer source: 0620_w25_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis

Answer points

3(a)

- $\rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$

- 2NaCl (only)

- (s) and (aq)

3(b)

- Mr $\text{BaCl}_2 = 208$

- $208 \rightarrow 0.1 = 20.8$

3(c)

- $0.100 \times 1.25 (\rightarrow 1000) = 80 \text{ (cm}^3\text{)}$

3(d)

- white

3(e)

- residue

3(f)

- rinsing the residue (with water)

- (mass was greater) because (dry) residue contains solid sodium chloride / NaCl adds to the mass (of residue)

3(g)

- precipitation

3(h)

- named soluble barium salt other than barium chloride e.g. (barium) nitrate

- named soluble sulfate salt other than sodium sulphate e.g. potassium (sulfate) / ammonium (sulfate)

- named insoluble barium salt other than barium sulfate e.g. (barium) carbonate

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*